

Learning Curves

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can read training-size learning curves.

I can describe variability across repeated development runs.

I can avoid unsupported guarantees from a trend.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

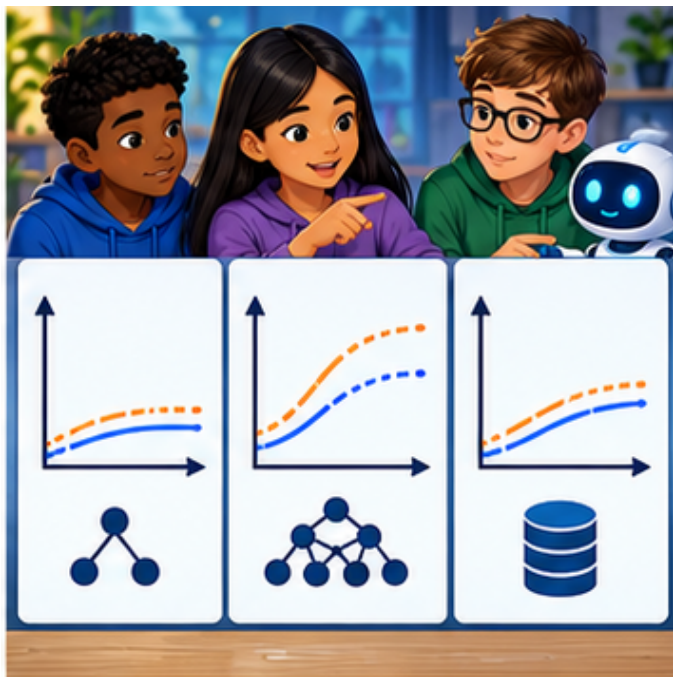
Without opening the old lesson, explain one key idea from Overfitting. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Learning curves plot performance against training size or effort and help distinguish underfitting, overfitting, and data limitation.



LEARNING TARGET Training sizes 20,40,80 yield validation accuracies 62%,70%,73%. Is doubling again guaranteed to yield 76%? Describe a defensible next experiment.

Builder name and experiment date:

Readiness for the connected explanation

What does validation loss add to training loss?

This week's end goal

By the end, I can explain learning curves using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → **Predict** → **Explain**

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use learning curves to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

A learning curve asks how performance changes as the amount of training evidence grows. Keep the validation protocol, model settings and metric comparable. Use subsets of training only; never add validation rows into the training subsets. Multiple repeats reveal sensitivity to sample selection or model randomness.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines learning curves and variability.

3. See how it works

ML MECHANISM

Input → **Model** → **Evidence**

IN

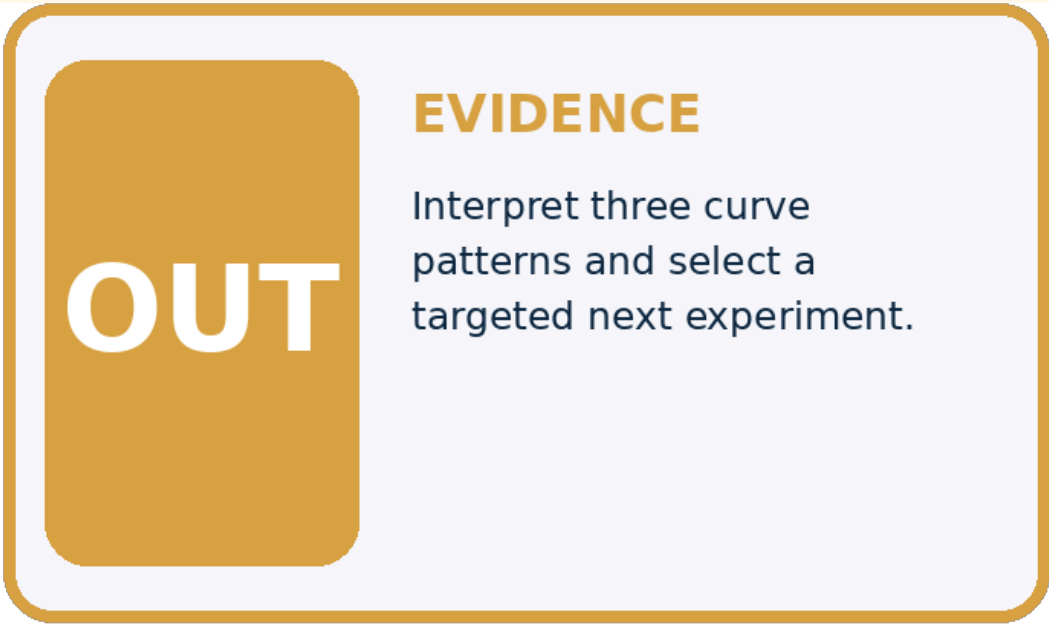
INPUT

Interpret three curve patterns and select a targeted next experiment.

ML

MODEL

Learning curves plot performance against training size or effort and help distinguish underfitting, overfitting, and data limitation.



OUT

EVIDENCE

Interpret three curve patterns and select a targeted next experiment.

QUESTION Interpret three curve patterns and select a targeted next experiment.

MODEL IDEA Learning curves plot performance against training size or effort and help distinguish underfitting, overfitting, and data limitation.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Both train and validation scores are low and close: likely underfitting. Train high, validation much lower: likely high variance or overfitting.

Draw a fresh data → model → prediction → evaluation trace:

A closer explanation

At training sizes 10, 20, 40, validation MAEs across three declared repeats are [7, 9, 8], [5, 7, 6], [4, 6, 5]. Means are 8, 6, 5 and ranges 2, 2, 2. Mean training MAEs are 1, 2, 3. Validation improves as training grows, while training error rises because fitting more varied examples can be harder.

Before continuing, underline the decision rule and circle the evidence that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim. Keep the distinction between an observed result and a proposed next step.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL CASE

Expected data from the intended setting



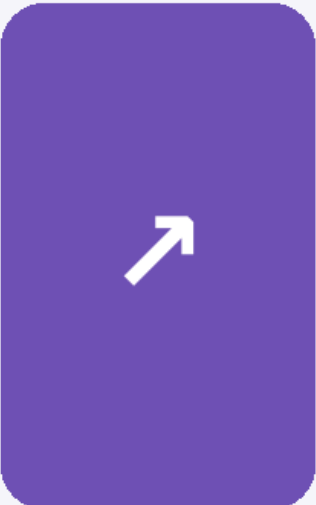
BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



SHIFTED CASE

New subgroup, device, condition, or time

NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: A30-plant experiment gives errors [4, 4, 4].

Prediction: Mean remains 4; this additional size does not improve the observed mean over 20 plants.

Reason: Learning curves are measured patterns, not a guarantee of improvement at every data size.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.



2

BIAS

systematic underfitting

3

VARIANCE

sensitivity to particular
samples

4

PLATEAU

little improvement from
added data

Term · Precise meaning · My ML example

Term: learning curve

Precise meaning: performance across training sizes

My ML example: _____

Term: bias

Precise meaning: systematic underfitting

My ML example: _____

Term: variance

Precise meaning: sensitivity to particular samples

My ML example: _____

Term: plateau

Precise meaning: little improvement from added data

My ML example: _____

MISCONCEPTION TO REPAIR Every learning curve that rises proves more data will solve the problem.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: Learning curve

Meaning: Performance plotted against training sample size

Word: Repeat

Meaning: Another run under a recorded procedure

Word: Variability

Meaning: How much results differ between runs

Word: Range

Meaning: Maximum minus minimum of observed values

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 28 • WORKED EVIDENCE

- 1 Both train and validation scores are low and close: likely underfitting.
- 2 Train high, validation much lower: likely high variance or overfitting.
- 3 Validation improves as data grows and gap narrows: more representative data may help.
- 4 Curves suggest hypotheses; confirm with controlled experiments and error slices.

STEP 1 Both train and validation scores are low and close: likely underfitting.

STEP 2 Train high, validation much lower: likely high variance or overfitting.

STEP 3 Validation improves as data grows and gap narrows: more representative data may help.

STEP 4 Curves suggest hypotheses; confirm with controlled experiments and error slices.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

Learning curves and variability at a glance

Part · Evidence or action · Interpretation

Part: 10 training rows

Evidence or action: Validation 7 9 8

Interpretation: Mean 8 range 2

Part: 20 training rows

Evidence or action: Validation 5 7 6

Interpretation: Mean 6 range 2

Part: 40 training rows

Evidence or action: Validation 4 6 5

Interpretation: Mean 5 range 2

The decreasing gap is encouraging in this experiment. It does not establish that doubling again will halve error. A range from three repeats is a descriptive spread, not a confidence interval or a guarantee that future results fall inside it. Repeated runs can share records and therefore are not independent studies.

Why can training error rise while validation error falls? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Learning Curves

Read a learning curve with repeated-run spread rather than drawing a certainty arrow.

The facts we will use

At 10 training plants validation errors across 3 runs are [6, 5, 7]; at 20 plants errors [4, 5, 3], on the same 10-case validation set.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Learning Curves

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Summarise first size

Mean 6 errors; range 5-7, corresponding to mean 40% accuracy.



2. Summarise larger size

Mean 4 errors; range 3-5, corresponding to mean 60% accuracy.



3. State bounded pattern

More training plants improved the average in these supplied runs; variability remains.

CHANGE ONE THING

A 30-plant experiment gives errors [4, 4, 4].

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Summarise first size

What happens and why: Mean 6 errors; range 5–7, corresponding to mean 40% accuracy.

Step: 2. Summarise larger size

What happens and why: Mean 4 errors; range 3–5, corresponding to mean 60% accuracy.

Step: 3. State bounded pattern

What happens and why: More training plants improved the average in these supplied runs; variability remains.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. More training plants improved the average in these supplied runs; variability remains. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

These toy runs do not establish that data quantity alone caused improvement; collection quality and conditions matter.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR

Tempting claim: Every learning curve that rises proves more data will solve the problem. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE Every learning curve that rises proves more data will solve the problem.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Collect additional data because it addresses a known coverage need, not merely to enlarge a count. When the curve stops improving, inspect labels, features and task suitability. Keep the final test reserved while planning these changes.

Why is an observed range not a confidence interval?

8. Practise together

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Record the actual subset identifiers and seeds for every repeat. If the validation population changes at the same time as training size, label that as a different experiment rather than attributing the difference only to more training data.

Practise with less help: Read both curves

Use workbook W2 for the connected example “Learning curves and variability”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Learning curves and variability. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Curve Clinic: match three graphs to diagnoses, challenge each diagnosis, and propose the smallest informative experiment.

1. Write the hypothesis and expected evidence before running.
2. Curve Clinic: match three graphs to diagnoses, challenge each diagnosis, and propose the smallest informative experiment.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

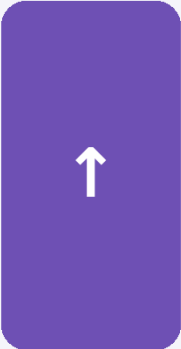
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY
1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: learning curve, bias, variance, plateau.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Every learning curve that rises proves more data will solve the problem.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.



RESPONSIBLE PLANT CLASSIFIER
Plot Plant Classifier train/validation performance at several training sizes and document the chosen next action.

THIS WEEK'S CHECKPOINT Plot Plant Classifier train/validation performance at several training sizes and document the chosen next action.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF
Interpret three curve patterns and select a targeted next experiment.

REQUIRED PROOF Interpret three curve patterns and select a targeted next experiment.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Feature Engineering.

13. Study a complete case

Week 28 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

With 20, 40, 80 training leaves, fixed-set validation accuracy is 60%, 70%, 78%.

The question

Interpret the curve and uncertainty.

Worked explanation

These runs improve with more data. Further representative data may help, but gains are not guaranteed. Controlled repeats reveal variability.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

More training data on the same validation set

Evidence or stage · Value or result

Evidence or stage: 20

Value or result: 60

Evidence or stage: 40

Value or result: 70

Evidence or stage: 80

Value or result: 78

With 20, 40, 80 training leaves, fixed-set validation accuracy is 60%, 70%, 78%.

Interpret the curve and uncertainty.

These runs improve with more data. Further representative data may help, but gains are not guaranteed. Controlled repeats reveal variability.

14. Try a new case independently

Independent case: Apply independently

At sizes 12 and 24, validation errors are [6, 8, 7] and [4, 7, 7]. Calculate means/ranges and a careful comparison.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 28 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

As training examples grow from 50 to 100 to 200, validation accuracy is 60%, 72%, 78%. What does the curve suggest, and what does it not guarantee?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Plot Plant Classifier train/validation performance at several training sizes and document the chosen next action.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Training sizes 20,40,80 yield validation accuracies 62%,70%,73%. Is doubling again guaranteed to yield 76%? Describe a defensible next experiment.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.